

Yash Prashant Mahajan

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EDUCATION

Rochester Institute of Technology

Master of Science in Computer Science

Relevant Coursework: Data structures & Algorithms, Big Data, Machine Learning, Artificial Intelligence

Aug 2023 - Dec 2025

Rochester, NY

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, HTML, CSS, C, C++.

Skills & Technologies: React.js, Node.js, FastAPI, Django, GraphQL, Spring Boot, Angular.js, Flask, Pandas, NumPy.

Databases & Cloud: MySQL, MongoDB, AWS (API Gateway, S3, Athena, Lambda, Glue, RDS, CodeBuild, AutoScaling, ELB), Azure.

Familiarity with: Linux, Git, Docker, Pytest, Junit, Terraform, GitHub Actions, Jenkins, Jira, CI/CD pipeline, Agile Methodology.

WORK EXPERIENCE

Software Engineer Co-op

May 2025 - Present

Syso Technologies

Boston, MA

- Developed a **React-Redux** dashboard to display battery optimization metrics for energy clients, enabling faster access to key data and **reducing load times by 80%** through **lazy loading** and code splitting.
- Engineered **RESTful API** endpoints using **Spring Boot** and **SQL queries** to retrieve pricing and reserve product data based on delivery dates, improving the accuracy and timeliness of financial reports used by stakeholders.
- Created 10+ **SQL AWS Athena views** for schedules and pricing data, integrated views through **Spring Boot** REST APIs to enable efficient querying and retrieval, improving reporting speed by 65%.
- Automated daily IESO financial reporting with **AWS Step Functions** and **AWS Lambda** querying IESO portal APIs, calculating revenue metrics, and writing results to **AWS DynamoDB**, reducing report generation time by 99%.
- Streamlined financial report generation by automating build processes with **Docker** and deploying infrastructure with **Terraform** through **GitHub Actions**, eliminating errors and reducing manual deployment effort by 99%.

Software Engineer

Jun 2022 - Jun 2023

Nihilent Technologies Pvt Ltd.

Pune, India

- Implemented an OCR-based **ETL** pipeline using **Python** and Azure Form Recognizer to extract key data from PDF invoices, transforming outputs into structured JSON and storing in AWS S3 for downstream analytics.
- Built an end-to-end fraud detection system in **AWS SageMaker** leveraging **AWS Rekognition** & rule-based logic to flag fraud bills.
- Processed 500+ daily invoices with 92% data-extraction accuracy, reducing invoice processing time by 63%.
- Automated ML workflows by orchestrating SageMaker jobs with AWS Step Functions triggered via **AWS EventBridge**, eliminating manual notebook runs and enabling fully automated, scalable model execution, reducing manual effort by 99%.

Software Engineer Intern

Jan 2022 - Jun 2022

Precise IT Solutions Pvt. Ltd

Pune, India

- Engineered responsive web interfaces using **React.js**, enhancing real-time interactivity & UX consistency across platforms.
- Designed **Spring Boot CRUD** APIs for product catalog, cart, and order endpoints, using Spring Data JPA for integration with the database ensuring structured data handling.
- Wrote **SQL** queries for data retrieval and reporting, joining product, cart, and order tables to validate API results, which improved data accuracy by 30%.

PROJECTS

AutoTradeIQ - AI Stock Navigator | React.js, Spring Boot, LangChain, OpenAI MCP, Dijkstra, Docker, CI/CD

- Architected a modular AI-driven trading platform leveraging **LangChain** and **MCP** to extract insights from SEC filings, earnings calls, and real-time market feeds, boosted stock prediction accuracy by 23% over the baseline.
- Designed **Spring Boot** microservices that orchestrate **LLM** analysis with financial datasets.
- Implemented **Dijkstra's algorithm** to predict swing high/low movements by optimizing risk-adjusted paths across 50+ equities.

Attire Avenue | React.js, SpringBoot, GraphQL, SQL, Docker, CI/CD, Jenkins

- Developed a full-stack e-commerce platform using **React.js** (frontend) and **Spring Boot** with **MySQL** (backend), following MVC architecture to manage and display 1,000+ products.
- Integrated **GraphQL** to streamline data querying and reduce payload size by 40%, improving frontend load times and responsiveness.
- Implemented user authentication, role-based access control, and secured API endpoints using **JWT**.

CERTIFICATIONS

- AWS Certified Solutions Architect - Associate